

Section 4.2 Mean Value Theorem

Suppose $f(x)$ is differentiable over $[a, b]$.

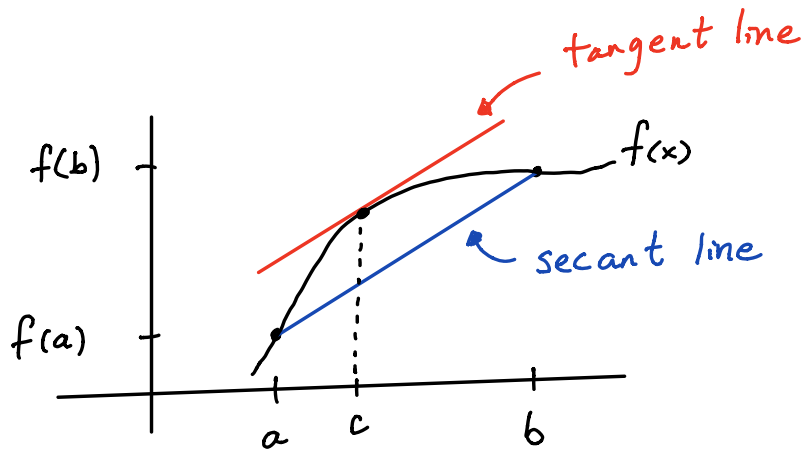
Then there exists a number c in (a, b)

so that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Slope of
tangent
line

↑
slope of
secant
line



The tangent line and secant line are parallel.

Ex ① Find the value of guaranteed by the Mean Value Theorem.

$$f(x) = -x^2 + 2x - 5 \quad \text{on } [0, 3]$$

$$f(3) = -3^2 + 2(3) - 5 = -8$$

$$f(0) = -0^2 + 2(0) - 5 = -5$$

$$\begin{array}{l} \text{Slope of} \\ \text{Secant} \\ \text{line} \end{array} = \frac{-8 - (-5)}{3 - 0} = \frac{-3}{3} = -1$$

$$\begin{array}{l} \text{Slope of} \\ \text{tangent} \\ \text{line} \end{array} = f'(x) = -2x + 2$$

Thus there is a value c in $(0, 3)$

$$\text{so that } -2c + 2 = -1$$

$$-2c = -3$$

$$c = \frac{3}{2}$$

Ex ② $f(x) = \sqrt{x}$ on $[1, 4]$

Find the value of c guaranteed by the mean value theorem.

$$f(4) = \sqrt{4} = 2$$

$$f(1) = \sqrt{1} = 1$$

$$\text{Slope of Secant Line} = \frac{2-1}{4-1} = \frac{1}{3}$$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$\frac{1}{2\sqrt{x}} = \frac{1}{3} \quad \text{then invert both sides}$$

$$2\sqrt{x} = 3$$

$$\sqrt{x} = \frac{3}{2}$$

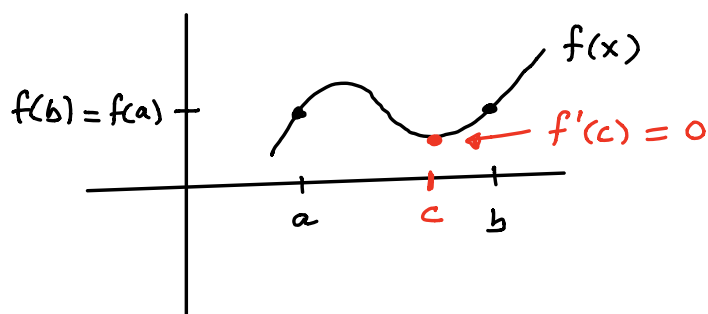
$$x = \frac{9}{4} \quad \text{or} \quad c = \frac{9}{4} \quad \text{or} \quad c = 1.75$$

which is in the interval $(1, 4)$

Rolle's Theorem

Suppose $f(x)$ is differentiable on $[a, b]$ and $f(a) = f(b)$.

Then there is a value c in (a, b) so that $f'(c) = 0$.



Ex ③ Verify Rolle's Theorem for

$$f(x) = e^x \cdot \sin x \quad \text{on } [0, \pi].$$

$$\begin{aligned} f(\pi) &= e^\pi \cdot \sin(\pi) = e^\pi \cdot 0 = 0 \\ f(0) &= e^0 \cdot \sin(0) = 1 \cdot 0 = 0 \end{aligned} \quad \begin{array}{l} \swarrow \\ \searrow \end{array} \quad f(b) = f(a)$$

Next find $f'(x)$ and solve $f'(x) = 0$

$$f'(x) = e^x \cdot \sin x + e^x \cdot \cos x$$

$$0 = e^x \cdot (\sin x + \cos x)$$

e^x will
never = 0

$$\sin x + \cos x = 0$$

$$(\sin x + \cos x)^2 = (0)^2$$

$$\sin^2 x + 2\sin x \cdot \cos x + \cos^2 x = 0$$

$$\sin^2 x + \cos^2 x + 2\sin x \cdot \cos x = 0$$

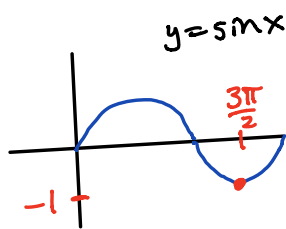
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$$1 + \sin(2x) = 0$$

$$\sin(2x) = -1$$

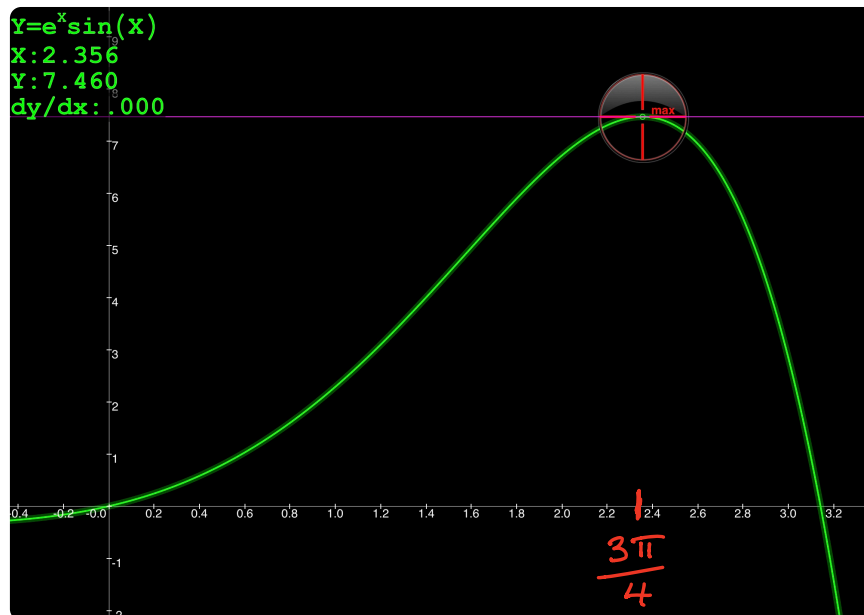
$$2x = \frac{3\pi}{2}$$

$$x = \frac{3\pi}{4} \text{ or } C = \frac{3\pi}{4}$$



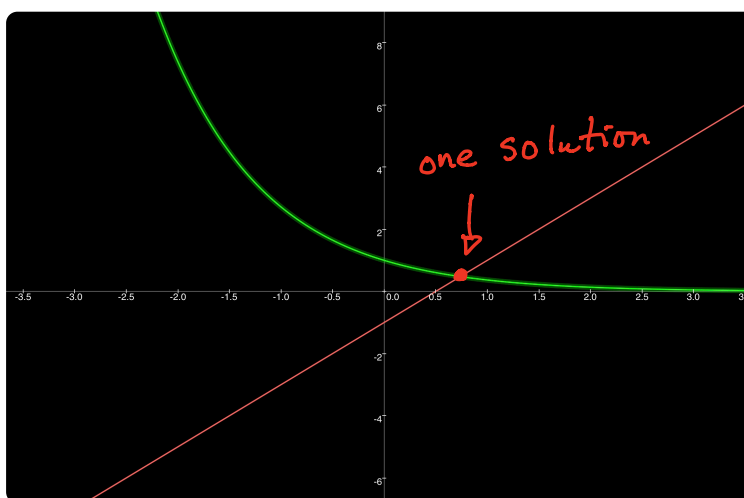
check the answer on the graph.

Notice that $\frac{3\pi}{4} = 2.356...$



Ex ④ Prove that $e^{-x} = 2x - 1$ has only one solution.

Graph it to see that there is at least one solution.



How can we show $e^{-x} = 2x - 1$ does NOT have two solutions?

First set it equal to zero.

$$e^{-x} - 2x + 1 = 0$$

Then let $f(x) = e^{-x} - 2x + 1$

And ASSume there are two

solutions $x = a$ and $x = b$

That mean $f(a) = 0$ and $f(b) = 0$.

By Rolle's Theorem, $f'(x) = 0$

MUST have a solution.

$$f(x) = e^{-x} - 2x + 1$$

$$f'(x) = -e^{-x} - 2$$

$$0 = -e^{-x} - 2$$

$$e^{-x} = -2 \quad \text{take ln of both sides.}$$

$$-x = \ln(-2)$$

↑ no solution!

This means that the assumption that there were 2 roots is false.

There can only be 1 solution!